

IDENTIFICATION

CODE : GI-3-S2-EC-ISC
ECTS : 3

HOURS

Cours : 6h
TD : 28h
TP : 12h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 48h
Travail personnel : 12h
Total : 60h

ASSESSMENT METHOD

IE (interrogation écrite) 80%
ES (évaluation en situation) 20%

TEACHING AIDS

Course and TD support
Exercices + Case studies

TEACHING LANGUAGE

French

CONTACT

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AIMS

This course belongs to teaching unit Supply chain management (GI-3-S2-UE-GECL) and contributes to the following skills :

- A1 Analyze a system (real or virtual) or a problem (Level 2)
- A2 Operate a model of a real or virtual system (Level 2)
- A5 Process data (Level 2)
- A6 Report on an analysis or on a scientific approach (Level 2)
- C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 2)
- C2 Modeling and designing an information, decision and production system, of goods and services (Level 2)
- C3 Evaluating, prototyping and simulating a system (Level 2)
- C8 Managing supply in connection with the planning and inventory management policy (Level 2)
- C9 Localizing and assigning the production, storage and transportation processes to different members of the supply chain (Level 2)
- C11 Understanding and evaluating a structure holistically, through socio-economic frames of reference (Level 1)
- B5 act responsibly in a complex world (Level 1)
- B6 position oneself, work, and evolve within a company or a socio-productive organization (Level 2)

By allowing the engineering students to work and be evaluated on the following knowledge :

- Forecasting methods, N&OP and their link with Demand management (A1, A2, C2, C8)
- Supply chain models (C2, C8, C9, B6)
- VBA syntax and event driven programming paradigm (C2)
- Bullwhip effect in the supply chain (C3)

To be able to :

- Identify the main functions of an industrial company, their operational logic, and their management methods. (C11, B6)
- Characterize the relationships of the company with its environment and within its supply chain, and the associated processes. (C2, C8, C9, B5, B6)
- Import and filter data using VBA syntax (A1, A2, A5, C1, C2)
- Manipulate event driven programming with VBA (user interaction and time trigger) (A2, A5, C2)
- Understand the problem of forecast management (A1, A2, C2, C8)
- Understand the stakes of the sales and operations planning (S&OP) and its link with (A1, A2, C2, C8)
- Select and apply a forecasting method adapted to a given context (A1, C1)
- To calculate the intermediate management balances, indicators, and ratios derived from the income statement and balance sheet. (A1, A6, C1, C11)

CONTENT

- Corporate structure
- Introduction to supply chain, SCOR Model
- Forecasting methods, S&OP, demand management
- case study: forecasting and statistical models & VBA programming for data import and filtering
- business finance (income statement, balance sheet), sustainable finance
- Beer game

BIBLIOGRAPHY

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- BAGLIN G., BRUEL O., GARREAU A., GREIF M., Van DELF C., Management Industriel et Logistique, Economica, 1996.
- GIARD V. Gestion de la production et des flux, 3e édition, Economica, 2003.
- JAVEL G. Organisation et gestion de la production, Masson, 1997.
- BOURBONNAIS R. et USUNIER J.C., Prévision des ventes, Economica 3rd éd. 2001.
- GOODRICH R., Applied statistical forecasting, Business Forecast System, 1992
- LEMANCQVIAUX P., La gestion prévisionnelle à court terme, Dunod, 1995